IP Addressing

Addressing

- Physical Addressing
  - Khan Sir Academy
  - (confirmly ~~not~~ reach)
  - → MAC Address (48 bits).
- Logical Addressing
  - (. - - . — - . 15/B, - - -, Bihar 799434)
  - (Confirm Reach).
    - IPV4
      - Public IP
      - Private IP.
    - IPV6

IPV4 (Internet Protocol Version 4)

- → 32 bit Logical Address
- → 4 Octet.
- → 0 - 255 (for every octet)
- → IP Address → Network/Net id
  - → Host Id.

$4 \times 8 = 32$

1 octet → 8 bit

⇒ 2^8

⇒ 256

0 255

IP

- Classfull IP-Address
  - Class-A
  - Class-B
  - Class-C
  - Class-D
  - Class-E
- Classless IP-Address.
  - ↓ CIDR Notations.

0 1111111 = 127

Class-A

8bit	8bit	8bit	8bit
0			

Netid = 8bit

Hostid = 24 bit

Min ✓ 0 | 0000000 . 0 . 0 . 0 ⇒ 0.0.0.0

Max 0 | 1111111 . 255 . 255 . 255 ⇒ 127.255.255.255

Netid = 8 bit

→ 0 | 0000000
0 | 1111111

$2^7 = 128$ Total Netid.

$2^7 - 2 = 128 - 2 = 126$ Usable Netid.

(1) { 0.0.0.0
0.255.255.255 }

(1) { 127.0.0.0
127.255.255.255 }

↳ Loop Back Address.

97 33
98...

62 | 43 | 5766

Hostid :→ 24 bit.

2^{24} → 16777216 (Total Host).

→ 16777216 − 2 = 16777214 (Usable Host).

Subnet Mask :→ 255.0.0.0 for class A. 18

⇓

We will find the Netid.

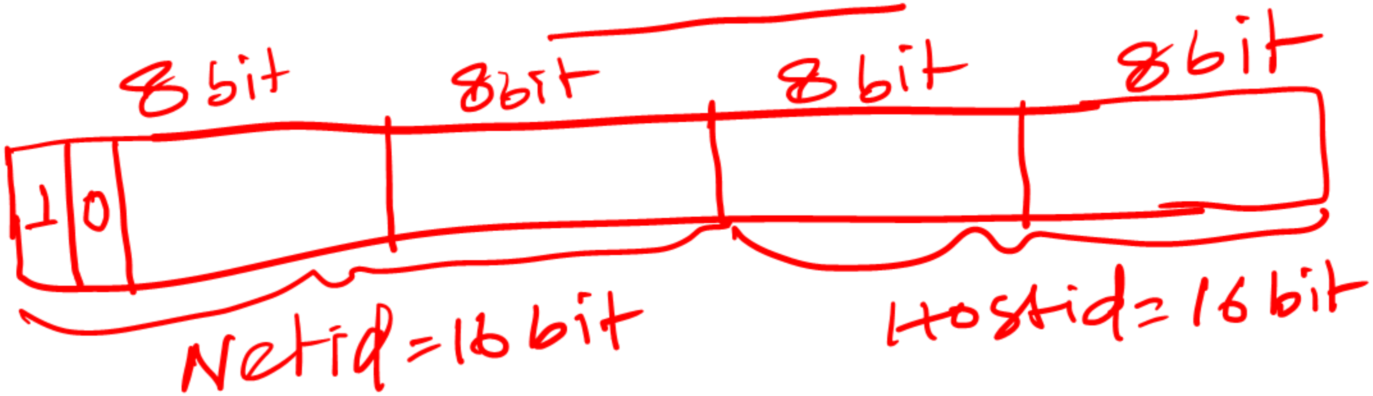
Bitwise AND

01101101.217.115.10
11111111.0.0.0

01101101.0.0.0 →

AND.

Class-B



10 | 000000 . 0 . 0 . 0 ⇒ 128.0.0.0

10 | 111111 . 255 . 255 . 255 ⇒ 191.255.255.255

Netid ⇒ 16 bit ⇒ (14 bit)

2^{14} ⇒ 16384 (Total no. of Netid)

⇒ 16384 − 2

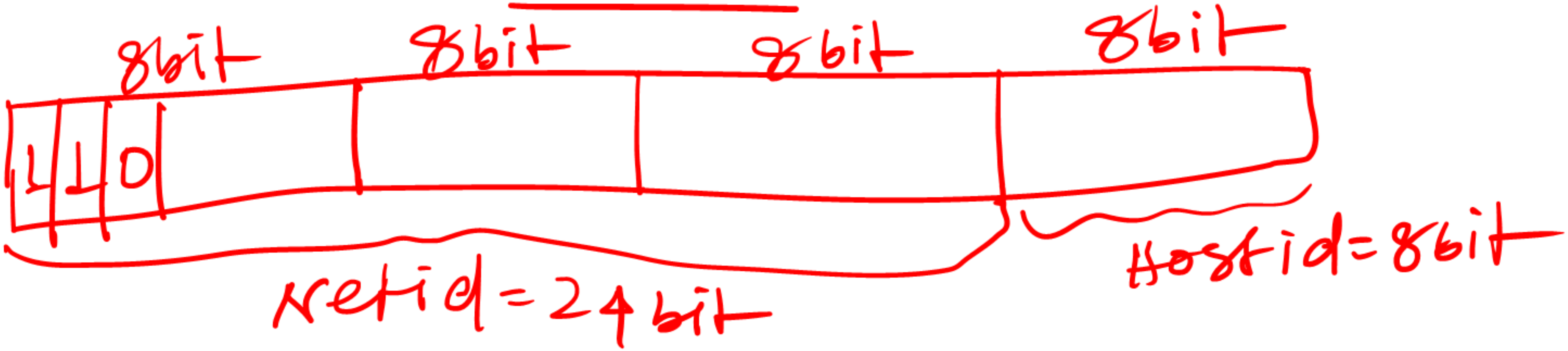
= 16382 (usable Netid).

Hostid ⇒ 16 bit ⇒ 2^{16} = 65536 (Total no. Hostid)

⇒ 65534 (usable).

Subnet Mask :→ $\boxed{255.255.0.0 \text{ for Class B}}$ /16

Class - C



110 | 00000 . 0 . 0 . 0 = 192.0.0.0

110 | 11111 . 255 . 255 . 255 ⇒ 223.255.255.255

Netid ⇒ 24 bit (21 bit)

⇒ 2^{21} = 2097152 (Total Netid)

= <u>2097150</u> (usable).

Host id = 8 bit
$= 2^8 = 255$ (Total)
$= 255 - 2 = \underline{253}$ (usable Host id)

subnet Mask ⇒ $\boxed{255.255.255.0 \text{ for class C}}$
/24

Class-D :→ 1110 | 0000 . 0 . 0 . 0 ⇒ 224 . 0 . 0 . 0
Multicast 1110 | 1111 . 255 . 255 . 255 ⇒ 239 . 255 . 255 . 255

Class-E :→ 1111 | 0000 . 0 . 0 . 0 ⇒ 240 . 0 . 0 . 0
Research 1111 | 1111 . 255 . 255 . 255 ⇒ 255 . 255 . 255 . 255
For Future use

Private IP (No Need to purchase)

- → class-A (10.0.0.0 to 10.255.255.255)
- → class-B (172.16.0.0 to 172.31.255.255)
- → class-C (192.168.0.0 to 192.168.255.255)

Public IP (You Need to purchase).

⊛ Find How many Host can connect in a specific Network (IP ⇒ 150.10.20.30)? Also find Netid and Broadcast IP?

Ans:→ IP ⇒ 150.10.20.30

that is a class-B IP.

so, Netid = 16 bit

Hostid = 16 bit.

subnet Mask = 255.255.0.0.

∴ Netid ⇒

$$\begin{array}{r} 150.10.20.30 \\ 255.255.0.0 \\ \hline 150.10.0.0 \end{array} \rightarrow \text{Netid.}$$

Broadcast IP ⇒ 150.10.255.255 → (Last IP of the Netid/IP).

No. of Host $\Rightarrow 2^{16}-2$

$= 65536-2$

$= 65534$